Assignment 2

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setting a working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading Libraries

```
library(dplyr)
library(ggplot2)
library(tidyverse)
library(sjmisc)
library(summarytools)
```

import file

```
mnp <- read.csv("mnpulate.csv")
head(mnp)
```

```
  Manufacturer    Model    Type Price MPG.city MPG.highway  Origin Wheelbase
1        Acura  Integra   Small  15.9       25          31 non-USA        97
2        Acura   Legend Midsize  33.9       18          25 non-USA       115
3         Audi   Legend Compact  25.9       20          26 non-USA       102
4         Audi  Century Midsize  30.8       19          25 non-USA       112
5          BMW Cavalier Midsize  36.9       22          30 non-USA       109
6        Buick  Century Midsize  14.2       22          31     USA       106
  Length Width Weight Horsepower
1    171    67   2705        140
2    195    71   3560        200
3    176    66   2810        172
4    193    74   3730        200
5    187    68   3050        208
6    189    69   3200        110
```

Task 1

filter cars to isolate wheelbase >110

```
wheel_filter <- mnp %>%
    filter(Wheelbase < 110)
head(wheel_filter)
```

```
  Manufacturer    Model    Type Price MPG.city MPG.highway  Origin Wheelbase
1        Acura  Integra   Small  15.9       25          31 non-USA        97
2         Audi   Legend Compact  25.9       20          26 non-USA       102
3          BMW Cavalier Midsize  36.9       22          30 non-USA       109
4        Buick  Century Midsize  14.2       22          31     USA       106
5    Chevrolet Cavalier Compact   8.5       28          36     USA        96
6        Acura  Integra   Small  15.9       25          31 non-USA        97
  Length Width Weight Horsepower
1    171    67   2705        140
2    176    66   2810        172
3    187    68   3050        208
4    189    69   3200        110
5    170    66   2410        110
6    171    67   2705        140
```

filter to include cars only of type sedan- USA

filter origin

```
originmnf <- mnp %>%
    filter(Origin == "USA")
head(originmnf)
```

```
  Manufacturer    Model    Type Price MPG.city MPG.highway Origin Wheelbase
1        Buick  Century Midsize  14.2       22          31    USA       106
2        Buick  LeSabre   Large  19.9       19          28    USA       111
3     Cadillac  DeVille   Large  34.4       16          25    USA       114
4    Chevrolet Cavalier Compact   8.5       28          36    USA        96
5    Chevrolet  Caprice   Large  18.8       17          26    USA       114
6        Buick  Century Midsize  14.2       22          31    USA       106
  Length Width Weight Horsepower
1    189    69   3200        110
2    201    74   3470        170
3    204    74   3730        200
4    170    66   2410        110
5    214    77   3960        170
6    189    69   3200        110
```

filter type

```
typec <- originmnf %>%
    filter(Type == "Sedan")
head(typec)
```

```
 [1] Manufacturer Model        Type         Price        MPG.city
 [6] MPG.highway  Origin       Wheelbase    Length       Width
[11] Weight       Horsepower
<0 rows> (or 0-length row.names)
```

Task 2

Selection of columns

continous columns

```
sel_cont <- mnp %>%
    select(Wheelbase, Length, Width, Weight)
head(sel_cont)
```

```
  Wheelbase Length Width Weight
1        97    171    67   2705
2       115    195    71   3560
3       102    176    66   2810
4       112    193    74   3730
5       109    187    68   3050
6       106    189    69   3200
```

categorical columns

```
sel_des <- mnp %>%
    select(Model, Origin, Type, Manufacturer)
head(sel_des)
```

```
     Model  Origin    Type Manufacturer
1  Integra non-USA   Small        Acura
2   Legend non-USA Midsize        Acura
3   Legend non-USA Compact         Audi
4  Century non-USA Midsize         Audi
5 Cavalier non-USA Midsize          BMW
6  Century     USA Midsize        Buick
```

task 3:

Creation or modification of variables

```
create <- mnp %>%
    mutate(weight_to_horsepower = Weight/Horsepower)
head(create)
```

```
  Manufacturer    Model    Type Price MPG.city MPG.highway  Origin Wheelbase
1        Acura  Integra   Small  15.9       25          31 non-USA        97
2        Acura   Legend Midsize  33.9       18          25 non-USA       115
3         Audi   Legend Compact  25.9       20          26 non-USA       102
4         Audi  Century Midsize  30.8       19          25 non-USA       112
5          BMW Cavalier Midsize  36.9       22          30 non-USA       109
6        Buick  Century Midsize  14.2       22          31     USA       106
  Length Width Weight Horsepower weight_to_horsepower
1    171    67   2705        140             19.32143
2    195    71   3560        200             17.80000
3    176    66   2810        172             16.33721
4    193    74   3730        200             18.65000
5    187    68   3050        208             14.66346
6    189    69   3200        110             29.09091
```

size class

```
size_class <- mnp %>%
    mutate(sized_class = ifelse(Wheelbase < 105, "Compact", ifelse(Wheelbase <= 110,
        "Midsize", "Large")))
head(size_class)
```

```
  Manufacturer    Model    Type Price MPG.city MPG.highway  Origin Wheelbase
1        Acura  Integra   Small  15.9       25          31 non-USA        97
2        Acura   Legend Midsize  33.9       18          25 non-USA       115
3         Audi   Legend Compact  25.9       20          26 non-USA       102
4         Audi  Century Midsize  30.8       19          25 non-USA       112
5          BMW Cavalier Midsize  36.9       22          30 non-USA       109
6        Buick  Century Midsize  14.2       22          31     USA       106
  Length Width Weight Horsepower sized_class
1    171    67   2705        140     Compact
2    195    71   3560        200       Large
3    176    66   2810        172     Compact
4    193    74   3730        200       Large
5    187    68   3050        208     Midsize
6    189    69   3200        110     Midsize
```

task 4:

continous descending

```
horse_desc <- mnp %>%
    arrange(desc(Horsepower))
head(horse_desc)
```

```
  Manufacturer    Model    Type Price MPG.city MPG.highway  Origin Wheelbase
1          BMW Cavalier Midsize  36.9       22          30 non-USA       109
2          BMW Cavalier Midsize  36.9       22          30 non-USA       109
3          BMW Cavalier Midsize  36.9       22          30 non-USA       109
4          BMW Cavalier Midsize  36.9       22          30 non-USA       109
5          BMW Cavalier Midsize  36.9       22          30 non-USA       109
6          BMW Cavalier Midsize  36.9       22          30 non-USA       109
  Length Width Weight Horsepower
1    187    68   3050        208
2    187    68   3050        208
3    187    68   3050        208
4    187    68   3050        208
5    187    68   3050        208
6    187    68   3050        208
```

ascending order to mpg.city

```
mpg_asc <- horse_desc %>%
    arrange(MPG.city)
head(mpg_asc)
```

```
  Manufacturer   Model  Type Price MPG.city MPG.highway Origin Wheelbase Length
1     Cadillac DeVille Large  34.4       16          25    USA       114    204
2     Cadillac DeVille Large  34.4       16          25    USA       114    204
3     Cadillac DeVille Large  34.4       16          25    USA       114    204
4     Cadillac DeVille Large  34.4       16          25    USA       114    204
5     Cadillac DeVille Large  34.4       16          25    USA       114    204
6     Cadillac DeVille Large  34.4       16          25    USA       114    204
  Width Weight Horsepower
1    74   3730        200
2    74   3730        200
3    74   3730        200
4    74   3730        200
5    74   3730        200
6    74   3730        200
```

sort cars by A-Z

```
a_z_sort <- mnp %>%
    group_by(Type) %>%
    arrange(Manufacturer)
head(a_z_sort)
```

```
# A tibble: 6 × 12
# Groups:   Type [2]
  Manufacturer Model   Type   Price MPG.city MPG.highway Origin Wheelbase Length
  <chr>        <chr>   <chr>  <dbl>    <int>       <int> <chr>      <int>  <int>
1 Acura        Integra Small   15.9       25          31 non-U…        97    171
2 Acura        Legend  Midsi…  33.9       18          25 non-U…       115    195
3 Acura        Integra Small   15.9       25          31 non-U…        97    171
4 Acura        Legend  Midsi…  33.9       18          25 non-U…       115    195
5 Acura        Integra Small   15.9       25          31 non-U…        97    171
6 Acura        Legend  Midsi…  33.9       18          25 non-U…       115    195
# i 3 more variables: Width <int>, Weight <int>, Horsepower <int>
```

SORTING BY TYPE NOW

```
sort_by_type <- a_z_sort %>%
    group_by(Type) %>%
    arrange(Manufacturer)
print(sort_by_type)
```

```
# A tibble: 140 × 12
# Groups:   Type [4]
   Manufacturer Model   Type  Price MPG.city MPG.highway Origin Wheelbase Length
   <chr>        <chr>   <chr> <dbl>    <int>       <int> <chr>      <int>  <int>
 1 Acura        Integra Small  15.9       25          31 non-U…        97    171
 2 Acura        Legend  Mids…  33.9       18          25 non-U…       115    195
 3 Acura        Integra Small  15.9       25          31 non-U…        97    171
 4 Acura        Legend  Mids…  33.9       18          25 non-U…       115    195
 5 Acura        Integra Small  15.9       25          31 non-U…        97    171
 6 Acura        Legend  Mids…  33.9       18          25 non-U…       115    195
 7 Acura        Integra Small  15.9       25          31 non-U…        97    171
 8 Acura        Legend  Mids…  33.9       18          25 non-U…       115    195
 9 Acura        Integra Small  15.9       25          31 non-U…        97    171
10 Acura        Legend  Mids…  33.9       18          25 non-U…       115    195
# i 130 more rows
# i 3 more variables: Width <int>, Weight <int>, Horsepower <int>
```

```
colSums(is.na(mnp))
```

```
Manufacturer        Model         Type        Price     MPG.city  MPG.highway
           0            0            0            0            0            0
      Origin    Wheelbase       Length        Width       Weight   Horsepower
           0            0            0            0            0            0
```

task 5:

compute summary statistics

mean, median, and standard deviation for continous variables

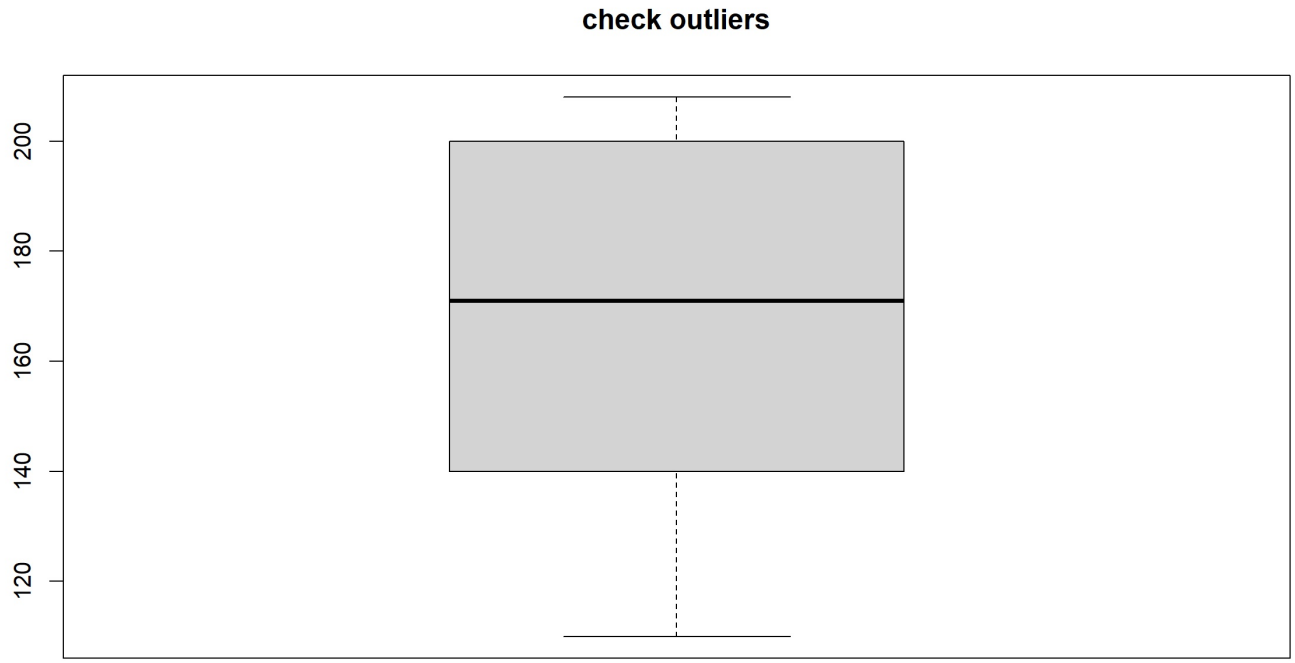
data checking for missing values

```
mv <- sum(is.na(mnp))
mv
```

```
[1] 0
```

CHECKING OUTLIERS 0n horsepower

```
boxplot(mnp$Horsepower, main = "check outliers")
```



perfoming mean, median and sd to horsepower

```
mean_value <- mean(mnp$Horsepower, na.rm = TRUE)
median_value <- median(mnp$Horsepower, na.rm = TRUE)
variance_value <- var(mnp$Horsepower, na.rm = TRUE)
sd_value <- sd(mnp$Horsepower, na.rm = TRUE)
print(paste("mean:", mean_value))
```

```
[1] "mean: 168"
```

```
print(paste("median:", median_value))
```

```
[1] "median: 171"
```

```
print(paste("variance:", variance_value))
```

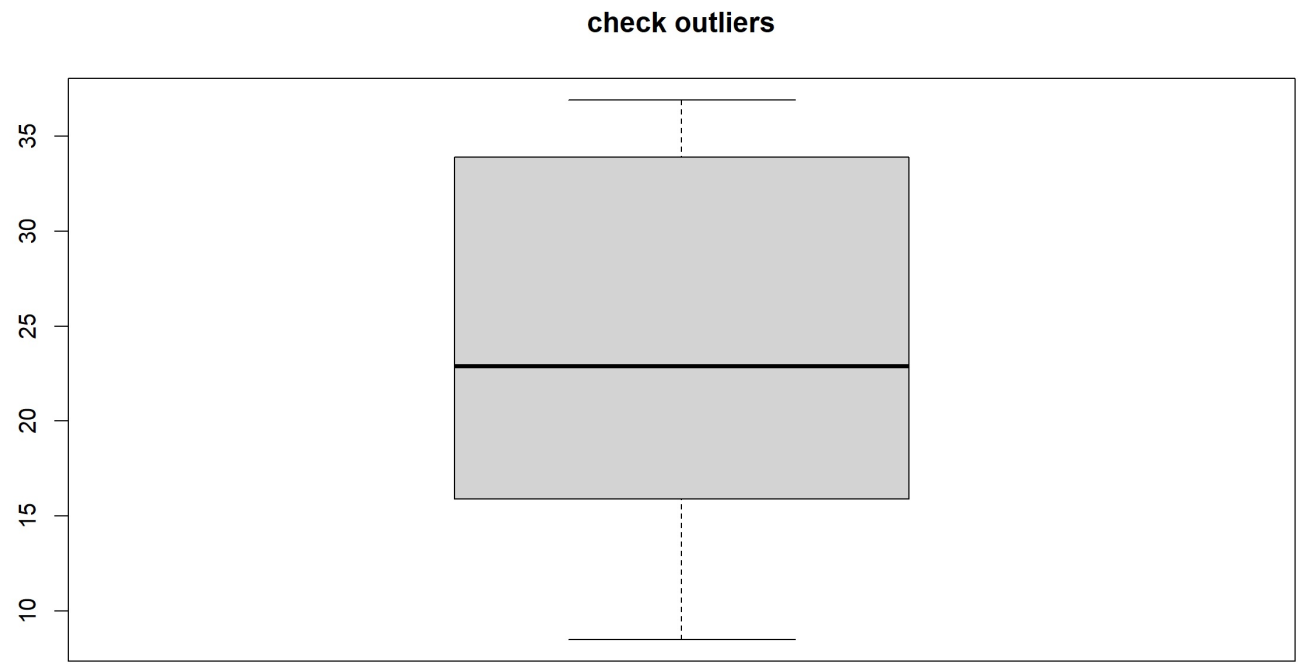
```
[1] "variance: 1229.58273381295"
```

```
print(paste("standard deviation:", sd_value))
```

```
[1] "standard deviation: 35.0654065114459"
```

CHECKING OUTLIERS on price

```
boxplot(mnp$Price, main = "check outliers")
```



perfoming mean, median and sd to price

```
mean_value <- mean(mnp$Price, na.rm = TRUE)
median_value <- median(mnp$Price, na.rm = TRUE)
variance_value <- var(mnp$Price, na.rm = TRUE)
sd_value <- sd(mnp$Price, na.rm = TRUE)
print(paste("mean:", mean_value))
```

```
[1] "mean: 23.92"
```

```
print(paste("median:", median_value))
```

```
[1] "median: 22.9"
```

```
print(paste("variance:", variance_value))
```

```
[1] "variance: 87.4361438848921"
```

```
print(paste("standard deviation:", sd_value))
```

```
[1] "standard deviation: 9.35072959104754"
```

```
colSums(is.na(mnp))
```

```
Manufacturer        Model         Type        Price     MPG.city  MPG.highway
           0            0            0            0            0            0
      Origin    Wheelbase       Length        Width       Weight   Horsepower
           0            0            0            0            0            0
```

summary statistics of horsepower and price

```
sam_tab <- mnp %>%
    group_by(Type) %>%
    summarise(count = n(), mean_power = mean(Horsepower), median_power = median(Horsepower),
        std_dv_power = sd(Horsepower), mean_price = mean(Price), median_price = median(Price),
        std_dv_price = sd(Price))
print(sam_tab)
```

```
# A tibble: 4 × 8
  Type    count mean_power median_power std_dv_power mean_price median_price
  <chr>   <int>      <dbl>        <dbl>        <dbl>      <dbl>        <dbl>
1 Compact    28       141          141         31.6       17.2         17.2
2 Large      42       180          170         14.3       24.4         19.9
3 Midsize    56       180.         200         40.6       29.0         32.4
4 Small      14       140          140          0         15.9         15.9
# i 1 more variable: std_dv_price <dbl>
```

group data by categories

summary statistics of type

```
sam_tab <- mnp %>%
    group_by(Type) %>%
    summarise(count = n(), mean_price = mean(Price), mean_mileage = median(MPG.city),
        )
print(sam_tab)
```

```
# A tibble: 4 × 4
  Type    count mean_price mean_mileage
  <chr>   <int>      <dbl>        <dbl>
1 Compact    28       17.2         24
2 Large      42       24.4         17
3 Midsize    56       29.0         20.5
4 Small      14       15.9         25
```